Design and Development of Detachable Continuum arm for soft applications

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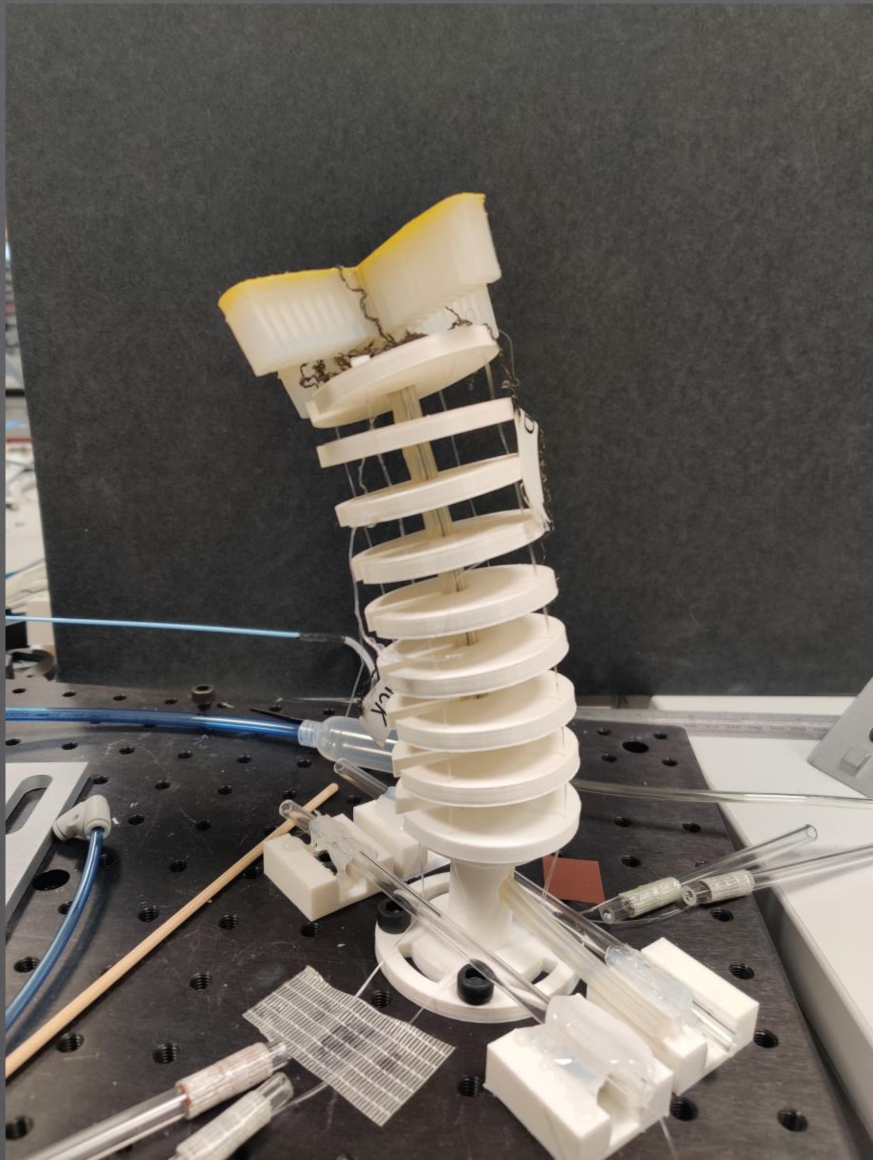
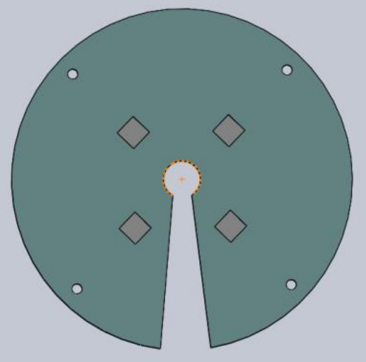
Overview

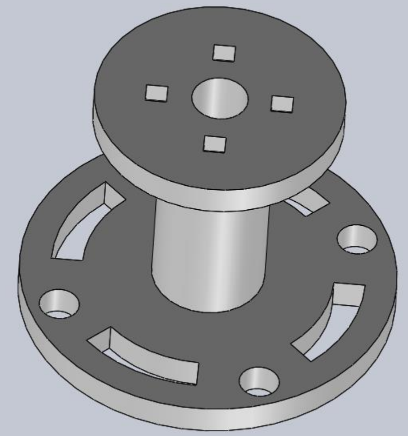
- Developed a novel continuum arm for soft applications.
- Utilized tendon-driven and pneumatic actuation systems.
- Eliminated the traditional backbone for improved flexibility.

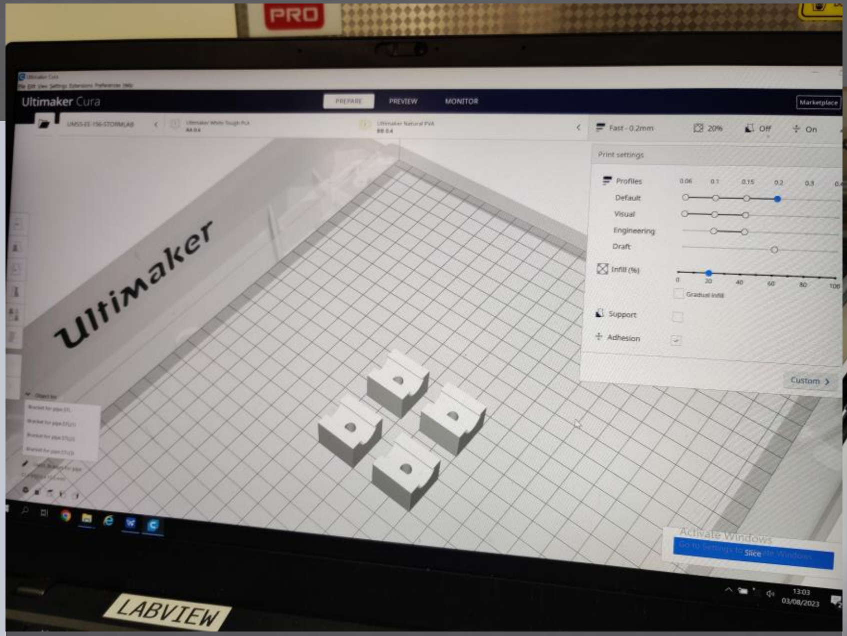


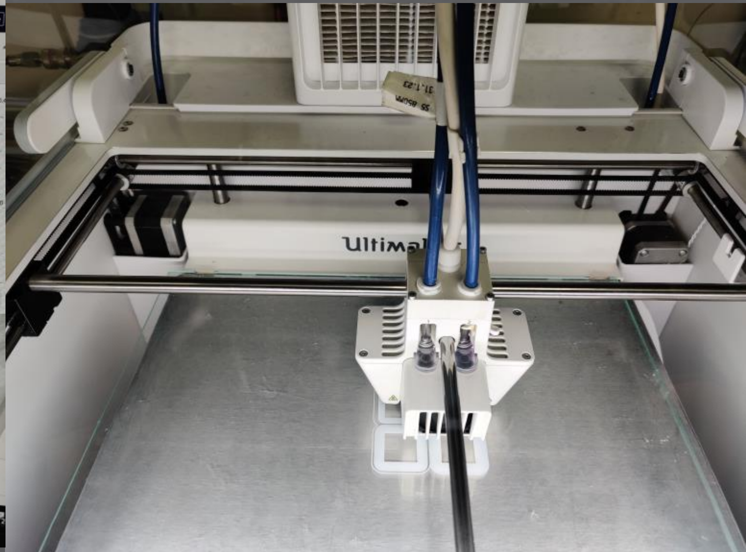
Design and Fabrication

- Designed a modular structure for easy adjustment and disassembly.
- 3D printed all arm components using PLA material for lightweight and flexibility.
- Utilized CAD software for precise design and prototyping.



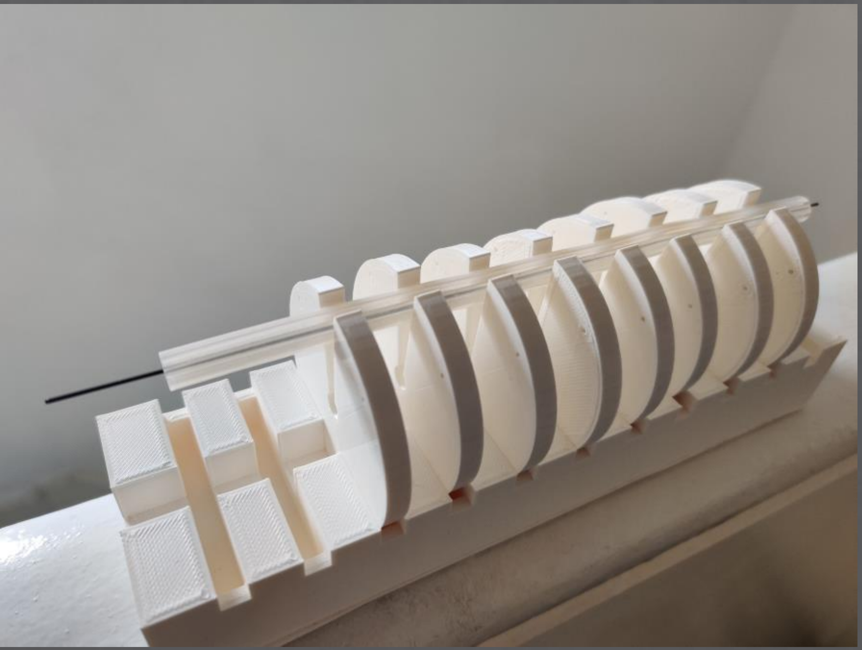


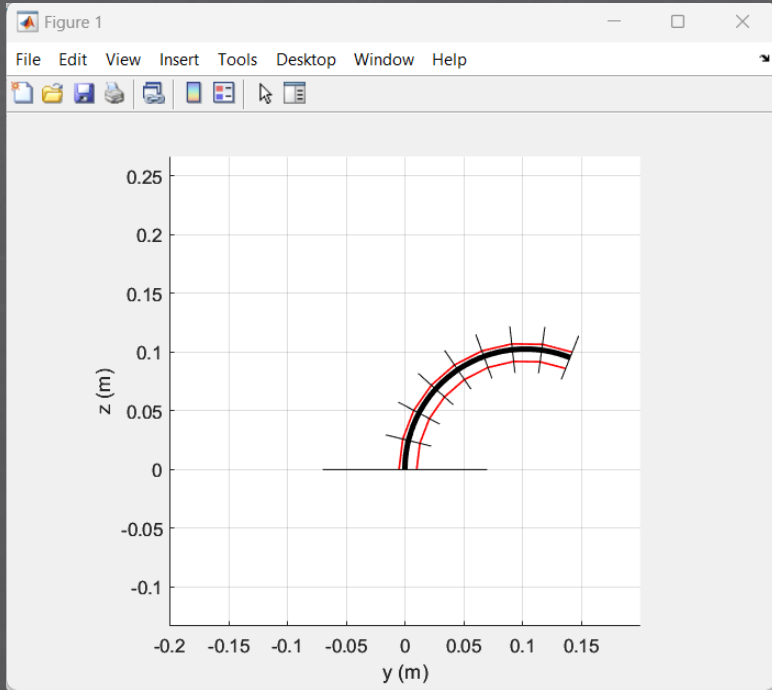




Testing and Validation

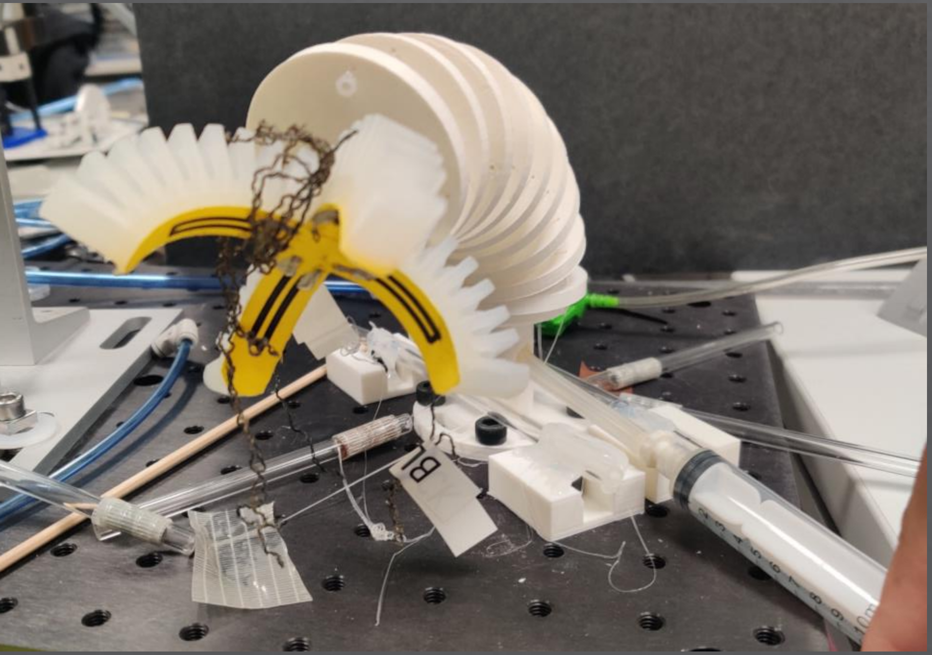
- Conducted rigorous testing to ensure optimal fit and functionality.
- Evaluated performance metrics such as reach, force, and dexterity.
- Made necessary adjustments based on test results.

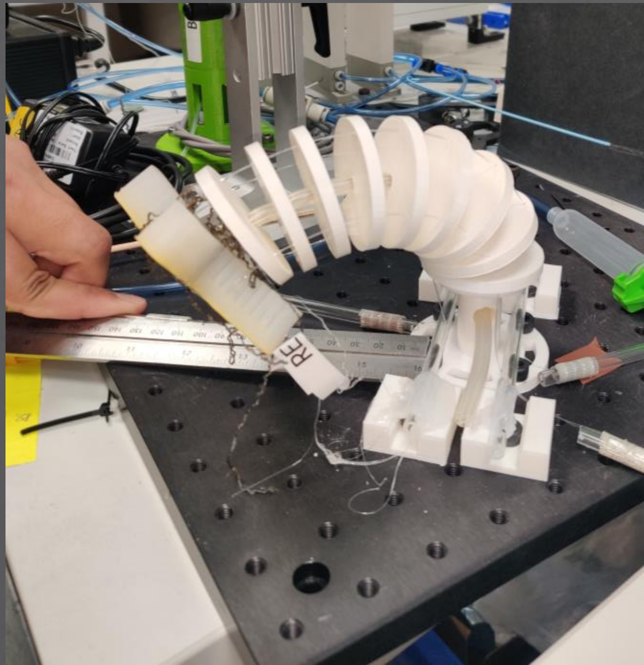


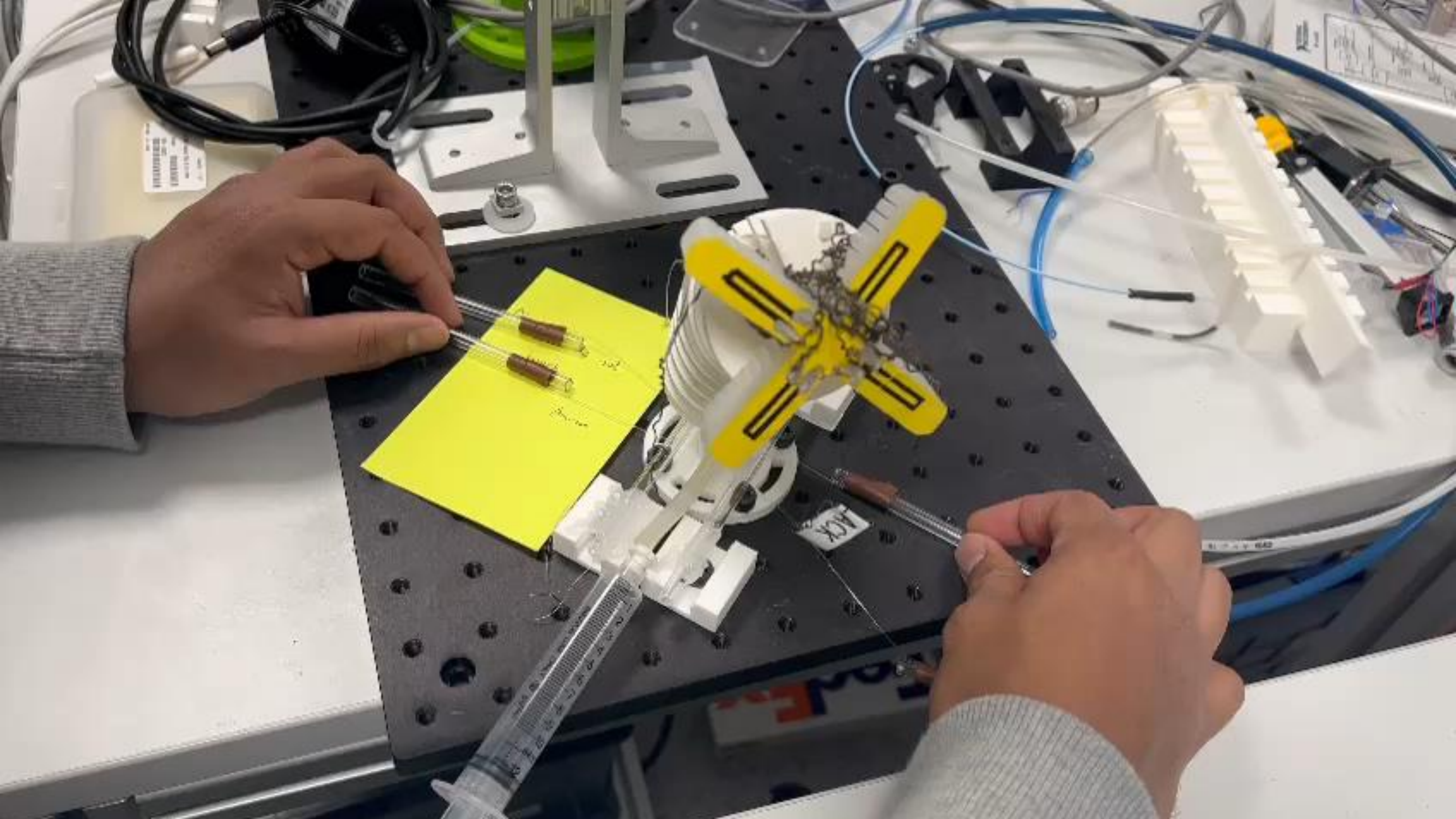


Key Achievements

- Successfully achieved project objectives and deliverables.
- Demonstrated innovative approach to continuum arm design.
- Contributed to advancements in soft robotics technology.








THANK YOU!